

Greening the HVAC System: A Logical Process



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There is a mantra that has been generated in the imaginations of most folks who want to build or remodel a home with green guidelines. The mantra goes: "I want my green home to have geothermal heating, bamboo floors, and solar panels on the roof, if I can afford them!"

This mantra tells me that the clients know very little about green building, and that I better start from the beginning when explaining green design concepts and processes, as I have been trying to do in the previous articles in this series. When it comes to greening your home for its heating and cooling needs, HVAC systems are no exception.

Green building is not about putting a bunch of pieces that have been marketed as "green" together to form a green house. There is a green logic required, and at no point in the development of a green home is this more critical than when considering the HVAC system. I think it would be useful to highlight this logic as it relates to this topic.

A valuable and logical consideration to start with is to design your green home's envelope to be efficient. When the design work for our first new-construction green home had begun, we wanted to create a house that would have such an energy-efficient building envelope that we would require very little energy to heat and cool it. With a goal of having a house of approximately 2,000 ft² or less, the original calculations called for a heating system that would provide up to 100,000 BTU/hr. Before the staff at Building Science Corporation was done, we had a house that



An AirCycler ventilation system invented by Armin Rudd monitors the operation of the air handler in this HVAC system, which is installed in one of our first green homes.



The same green house uses a single air handler to provide space heating and space cooling.

would require only 28,000 BTU/hr. If you can build a house that will heat for so little, it is beyond any reasonable logic to build otherwise, assuming the materials you use also meet green building standards.

The second piece of green building HVAC logic has to do with house size. I was asked to consult on a project of several town homes, each of which would be 4,000 ft² or more. I refused, because creating a home of this size defies green logic. Creating an HVAC system for such a house defies green logic. In a time when there are millions of people struggling to pay continually escalating fuel prices, allowing people to

build such huge homes is penalizing the people least able to pay.

I have a friend, an internationally known city planner, who has for years advocated for a concept he calls equity planning. He believes that we should not permit such oversized housing to be part of any plan, but since it is likely to continue, homeowners should have to find a way to pay for the additional demand they are putting on the system. Creating high-efficiency equipment to heat, cool, and ventilate such huge houses is like putting a high-efficiency motor system in a Hummer. It is simply inappropriate. We should not be wasting our time on such efforts when the need for efficient HVAC systems is so great in smaller housing.

The third piece of logic has to do with creating and maintaining incentives to encourage appropriate building. Developers in Cleveland can obtain land very cheaply from a city-owned land bank to build residential housing, and they can offer their buyers tax abatement for 15 years. It is time for the city to demand that anyone using such measures be required to build green, and a basic standard should be set. This standard should include the size of the structure, the optimum building envelope, and the appropriate HVAC measures. The housing of lower-income folks should also be green, since green, energy-efficient homes are more affordable.

The fourth piece of logic concerns the design of HVAC systems. Poor design and poorly located heating/cooling systems cause durability and comfort problems, as illustrated in the following two houses that I was asked to troubleshoot. In the first house, the heating/cooling system and many ducts were installed in the attic, and the residents were baffled when water leaked into the house from snowmelt on the roof. The snowmelt was caused by residual heat from the units, which would have been a helpful source of heat if kept inside the building envelope. In the second house, the heating/cooling system was located at one end of the basement so the sound of the air handler would not interfere with the entertainment center at the other end of the basement. Yet it was a mystery to the residents and the HVAC contractor why very little heat got to this room at the end of an enormously long heat run. Centering a heating system in the middle of the basement reduces duct run length and makes it easier to balance the system.

The design of the HVAC system often gets seriously considered only after the floor plan for the house is created, and then the trick is trying to incorporate the system into the floor plan. This process often results in convoluted duct runs, making it nearly impossible to balance a forced-air system effectively. The HVAC design and the house design must be developed with a sensitivity to the demands each will place on the final product. I have been in too many new homes that are not comfortable to live in and are not achieving the savings anticipated because the HVAC system is in tension with the layout and occupant functioning in the home.

In addition, logical design dictates that we simplify the HVAC systems we create. We have made huge strides in creating more energy-efficient heating and cooling equipment and the new ASHRAE 62.2 guidelines will soon have us incorporating full ventilation measures in all the homes we build, and eventually (let's hope) in all homes we rehab as well. But we are likely to end up with a heating/cooling unit or units,

a water heating system of some kind, and a ventilation system.

In our first green homes, Armin Rudd, the HVAC engineer, helped us incorporate everything in a single system. A hot-water tank became the primary heat source for the house as it fed hot water to a coil in an air handler that provided space heating when a fan blew air over it; the same air handler



The water heater serves as the heat source for all water services and space heating.

provided space cooling when the fan blew over cooling coils during cooling season. Fresh air was brought into the return side of the air handler through a duct to the exterior. An AirCycler ventilation system invented by Armin monitored the operation of the air handler (whether in heating or in cooling mode). If the air handler did not run for a set (programmable) length of time, the AirCycler would activate the fan, thereby both providing fresh air (ventilation) and mixing the air throughout the house (increasing thermal comfort). The same hot-water tank provided the water for cooking, clothes washing, dishwashing, and bathing.

In the first winter (2001–02), the total cost for heating (all hot water needed for heating and other needs) for six months (October through March) was \$379, and the house was comfortable

in each room. We used ductwork to deliver heating/cooling/ventilation air, but used transfer grilles in each room to provide the return air pathway from the room to the hallway even when the door was closed. This system balances the air going into and leaving each room, as documented by a pressure difference between the room and the hallway of less than 3%. Because we air sealed the building envelope and sealed the air ducts, and because there were no ducts in exterior walls, there was no leakage from the ducts to the exterior. A Canadian company is producing a system called Lifebreath that incorporates all of the features we included, including a heat recovery ventilation system. What we produced is not the last word by any means, but it is consistent with the logic of simplification and consolidation.

A final piece of logic is that our HVAC systems, which generally use more natural resources to operate than anything else in our homes, must be designed so they use fewer and fewer resources, but also have the least impact on both the interior and the exterior environment. Sealed-combustion or power-vented equipment is helping protect the interior environment, and the efficiency of equipment is greatly reducing the amount of pollutants entering the exterior environment.

As a lurker on listservs where persons far more knowledgeable than I are debating the fine points of HVAC systems (sometimes like theologians debating how many angels can stand on the head of a pin), I am continually looking for products created to provide comfort that also meet the logic tests I have just described. When they do not, I just keep looking and encourage folks who talk to me about building a green home to keep looking as well. Keep looking for ways to reduce the heating/cooling requirements, keep looking for what is basically required, keep looking for heating/cooling delivery systems that are “simply” effective, and keep looking for a system that is carefully integrated into the entire living space so comfort is brought evenly to all of it.

